

John T. Griesmer

Curriculum Vitae - September 2026

CONTACT INFORMATION

Home address:
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USA

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EDUCATION

The Ohio State University Columbus, OH

Ph.D. Mathematics, June 2009

- Dissertation: “Ergodic averages, correlation sequences, and sumsets”
- Advisor: **Vitaly Bergelson**

The Ohio State University

M.S. in Mathematics, December 2003

Miami University

B.S. in Mathematics, May 2002 *Magna Cum Laude*

CURRENT POSITION

Teaching Associate Professor
Department of Applied Mathematics and Statistics
Colorado School of Mines, Golden, CO USA

August 2021 – Present

FORMER POSITION

Adjunct Faculty
Department of Applied Mathematics and Statistics
Colorado School of Mines, Golden, CO USA

August 2016 – May 2021

PREVIOUS POSITIONS

Developer/Manager, Elephant Head Software, Denver CO, USA

May 2015 – May 2016

Tutor (self employed), Denver, CO, USA

January 2015 – April 2015

Lecturer, Department of Mathematics
University of Denver, Denver, CO, USA

September 2012 – November 2014

Lecturer, Department of Mathematics
The Ohio State University, Columbus, OH, USA

September 2011 – June 2012

Postdoctoral Fellow, Department of Mathematics
University of British Columbia, BC, Canada

September 2009 – August 2011

CITIZENSHIP

United States of America

AWARDS	Colorado School of Mines Earth & Society Programs Portfolio	
	Outstanding Teaching Assistant Professor	2024-2025
TEACHING	Colorado School of Mines	
	Math 112 - Calculus II	Fall 2019, '20
	Math 112 - Calculus II (2 sections, coordinating 11 sections)	Fall 2021
	Math 225 - Differential Equations	Fall 2022, '25, '26, Spring '26
	Math 213 - Multivariable Calculus	Fall 2016, '17, '18,'24, Spring '17, '18, '19
	Math 300 - Foundations of Advanced Mathematics	Spring 2019, Fall 2019
	Math 301 - Introduction to Analysis	Spring 2018, '22, '23, '24, '25, '26, Fall '21
	Math 307 - Introduction to Scientific Computing	Fall 2018, '22, '23, '24, '26 Spring 2022, '23, '24
	Math 332 - Introduction to Linear Algebra	Spring 2017
	Math 334 - Introduction to Probability	Spring 2025
	Math 599 - Analysis and Additive Combinatorics (independent study)	Spring 2025
STUDENT WORK DIRECTED	Rigidity sequences in dynamical systems	
	Advised Colorado School of Mines undergraduate Cash Cherry in a project to conduct original research on rigidity sequences for groups of measure preserving transformations. This project was funded as a MURF (Mines Undergraduate Research Fellowship). Cash solved an open problem posed by Ethan Ackelsberg and published the solution as a single-authored article in <i>Colloquium Mathematicum</i> , a pure mathematics journal well-known in abstract harmonic analysis:	
	C. CHERRY, Rigid-recurrent sequences for actions of finite exponent groups, <i>Colloq. Math.</i> 178 (2025), no. 2, 193–199, DOI:10.4064/cm9322-5-2025	
DEPARTMENT SERVICE	AMS Teaching Postdoc Search Committee Member	January 2026 - March 2026
	AMS Awards Committee	August 2025 - present
	Propose faculty, staff, and student award nominations.	
	AMS Teaching Faculty Search Committee Member	Jan - Mar 2025 (search canceled)

DEPARTMENT
SERVICE
(CONTINUED)

Faculty advisor for

Mines Mathematics and Computing Collaborative June 2022-May 2024, May 2025 - present

Perform administrative duties for the MMCC, a Colorado School of Mines student organization dedicated to increasing opportunity for mathematics, statistics, and computing involvement for underrepresented groups through outreach.

Outreach Committee Member

Developed a system for equitably distributing outreach duties among department members

Fall 2023 - Spring 2024

Linear Algebra Curriculum Committee Member

Identify learning outcomes to standardize Math 332 (Linear Algebra) delivery on campus

Fall 2023 - Spring 2024

Undergraduate Committee Member

Colorado School of Mines Dept. of Applied Mathematics and Statistics

August 2021-present

Undergraduate Core Curriculum Subcommittee Member

Propose reforms to Applied Mathematics and Statistics core curriculum

Sept. 2021-March 2022

Faculty advisor for SIAM Student Chapter (Math Club)

Assist student leadership in organizing events and administrative duties

August 2021-present

Putnam club advisor

Lead Putnam Exam preparation and general problem-solving sessions, proctor the Putnam Competition at Mines

August 2021-present

PUBLICATIONS

1. *Kneser- and Jin-type inverse theorems in discrete abelian groups* (2026), submitted <https://arxiv.org/abs/2602.19014v1>
2. *Discrete sumsets with one large summand*, Form of Mathematics, Sigma **14** (2026) [article link](#)
3. *Separating measurable recurrence from strong recurrence via rigidity sequences*, Discrete and Continuous Dynamical Systems (2025), doi.org/10.3934/dcds.2025024
4. *Separating topological recurrence from measurable recurrence: exposition and extension of Kriz's example*, Australasian Journal of Combinatorics (2025), https://ajc.maths.uq.edu.au/pdf/91/ajc_v91_p001.pdf
5. (with Pierre-Yves Bienvenu, Anh Le, and Thái Hoàng Lê) *Intersective sets for sparse sets of integers*, Ergodic Theory & Dynamical Systems (2024), doi.org/10.1017/etds.2024.73
6. *A set of 2-recurrence whose perfect squares do not form a set of measurable recurrence*, Ergodic Theory & Dynamical Systems (2024), doi.org/10.1017/etds.2023.51
7. (with Anh Le and Thái Hoàng Lê) *Bohr sets in sumsets II: countable abelian groups*, Forum Math. Sigma **11** (2023), Paper No. e57, 34 pp. doi.org/10.1017/fms.2023.49
8. *Special cases and equivalent forms of Katznelson's problem on recurrence*, Monatshefte für Mathematik (2023), doi.org/10.1007/s00605-022-01770-6
9. *Bohr neighborhoods in generalized difference sets*, The Electronic Journal of Combinatorics (2022), doi.org/10.37236/10622

PUBLICATIONS
(CONTINUED)

10. *Separating Bohr denseness from measurable recurrence*,
Discrete Analysis (2021), arxiv.org/pdf/2002.06994
11. *Semicontinuity of structure for small sumsets in compact abelian groups*,
Discrete Analysis (2019), arxiv.org/pdf/1807.01694
12. (with Michael Björklund) *Bohr sets in triple products of large sets in amenable groups*,
Journal of Fourier Analysis and Applications (2019) doi:10.1007/s00041-018-9615-5
13. *Recurrence, rigidity, and popular differences*,
Ergodic Theory and Dynamical Systems (2019) doi:10.1017/etds.2017.71
14. *An inverse theorem: when the measure of the sumset is the sum of the measures in a locally compact abelian group*,
Transactions of the American Mathematical Society **366** (2014) 1797-1827.
doi:10.1090/S0002-9947-2013-06022-9
15. *Small-sum pairs for upper Banach density in countable abelian groups*,
Advances in Mathematics, **246** (2013) 220-264. doi:10.1016/j.aim.2013.06.005
16. *Sumsets of dense sets and sparse sets*,
Israel Journal of Mathematics, **190** (2012), 229-252. doi:10.1007/s11856-012-0008-1
17. *On the multiplicity of jigsawed bases in compact and countably compact spaces*,
with Zoltan T. Balogh, Topology and its Applications **130** (2003), 65-73.
doi:10.1016/S0166-8641(02)00198-0

INVITED TALKS

- Graduate Workshop in Logic and Combinatorics,**
University of Illinois Chicago August 10 - 14 2026
Delivered a week-long graduate tutorial on additive combinatorics.
- American Mathematical Society Joint Meetings**
Special session on Ergodic Theory and its Connections
with Combinatorics and Harmonic Analysis January 2026
Discrete sumsets with one large summand
- Colorado State University**
Rocky Mountain Algebraic Combinatorics Seminar (2 talks) September 2025
1. *Kneser's Theorem for asymptotic density*
2. *Generalizing Kneser's theorem*
- Colorado State University**
Rocky Mountain Algebraic Combinatorics Seminar February 2025
Configurations in dense sets, positive definite functions, and Fourier transforms
- University of Florida**
South Eastern Logic Symposium 2024 March 2024
Ultraproduct correspondence principles for sumsets
- American Mathematical Society Joint Meetings**
Special session on Ergodic theory, symbolic dynamics, and related topics January 2024
A hierarchy of rigidity properties for measure preserving systems
- University of Mississippi Combinatorics Seminar** September 2023
Chromatic Number and Independence Number of Cayley Graphs on abelian groups

INVITED TALKS
(CONTINUED)

Nilpotent structures in topological dynamics, ergodic theory, and combinatorics Będlewo, Poland <i>Separating recurrence properties</i>	June 2023
AMS Southeastern Sectional Meeting Special Session on Discrete Analysis <i>Special cases and equivalent forms of Katznelson's problem on recurrence</i>	March 2023
Colorado State University Rocky Mountain Algebraic Combinatorics Seminar <i>Structure in sumsets</i>	March 2023
University of Wyoming Algebra, Combinatorics, and Number Theory seminar <i>A brief history of ergodic theory and additive combinatorics</i>	October 2022
Chalmers University Analysis Seminar <i>Rigidity Sequences for Measure Preserving Transformations</i>	June 2022
University of Mississippi Algebra and Number Theory Seminar <i>Inverse theorems for sumsets in additive combinatorics</i>	March 2022
AMS Spring Eastern Virtual Sectional Meeting Crossroads: Ergodic Theory, Harmonic Analysis, and Combinatorics <i>Special cases and equivalent forms of Katznelson's problem on recurrence</i>	March 2022
The Ohio State University Ergodic Theory Seminar (remote) <i>Rigidity sequences for measure preserving transformations</i>	March 2022
The 9th Visegrad Conference on Dynamical Systems, Prague 2021 (remote) <i>Problems and recent results in single recurrence</i>	June 2021
University of Lille Functional Analysis Seminar (remote) <i>Single recurrence: recent results, open problems, and mysterious examples</i>	April 2021
University of Warwick Ergodic Theory and Dynamical Systems Seminar University of Warwick, Coventry, UK (remote) <i>Single recurrence: recent results, open problems, and mysterious examples</i>	October 2020
Hebrew University Dynamical Systems Seminar (remote) Hebrew University, Jerusalem, Israel <i>Separating Bohr denseness from measurable recurrence</i>	June 2020
Dynamics and Number Theory Conference University of Sydney, Sydney, Australia <i>Recurrence and rigidity in measurable dynamics</i>	June 2019
University of New South Wales Number Theory Seminar <i>Inverse theorems in additive combinatorics</i>	June 2019
Max Dehn Seminar University of Utah, Salt Lake City, USA <i>Recurrence and rigidity</i>	March 2019